MultiSenseGolf: A Multimodal Sensor-Based Dataset for Golf Swing Motion and Shot Performance Analysis

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ABSTRACT

MultiSenseGolf is a multimodal dataset for golf swing motion and shot performance analysis collected using wearable and environmental sensors. It contains 1,557 swing records from 24 participants spanning Beginner, Intermediate, and Professional skill levels. The released modalities include full-body joint kinematic estimates from an inertial motion capture (IMC) system, bilateral foot pressure streams, first-person view (FPV) video, gaze coordinates, acceleration and angular velocity from a head-mounted inertial measurement unit (IMU), front-view RGB-D video, and shot outcome data recorded using an optical launch monitor. To support swing-level analyses, we conducted post-hoc data processing by annotating the start and end boundaries of each swing, correcting the observed timestamp offsets between recording systems using different device clocks, and storing the processed streams as swing-level records. Technical validation comprises four analyses: a mathematical sensitivity analysis of forward kinematics (FK)-derived joint positions under assumed orientation errors, a statistical analysis of carry distance distributions across participant attributes, a characterization of observed cross-device timestamp offsets, their within-session trends, and cross-modal timing variability, and a shot outcome prediction benchmark across five input conditions based on different modalities.

Background & Summary

A golf swing is a high-speed whole-body movement in which multiple joints and body segments must coordinate sequentially within a short time interval[1]. Due to the sensitivity of this coordination, even subtle deviations in timing or body rotation can substantially degrade performance-critical outcomes such as ball flight[2]. As a result, achieving a precise and consistent swing requires continuous, repetitive practice across all skill levels, from novices to professionals[1,3]. During self-guided practice, golfers primarily rely on intrinsic feedback sources, such as the tactile sensation of ball contact and the proprioception of body segments, to evaluate their swing execution[4,5]. However, the rapid dynamics of the golf swing often limit the effectiveness of intrinsic feedback, as sensory cues tend to merge into an overall bodily sensation[6]. Moreover, golf swing performance is influenced not only by joint kinematics, but also by diverse motion-related cues such as ground–body interaction (e.g., weight transfer and balance)[7,8], visual attention[9], and head stability[10], which further complicates self-guided training.

Accordingly, augmented feedback is widely used in golf coaching and training to complement information that intrinsic feedback alone cannot provide. Augmented feedback is information delivered from external sources and can be provided through multiple modalities, such as visual, auditory, and haptic cues[11]. In golf, instruction and correction from a coach are common, but cost and accessibility constraints make them difficult to rely on during self-practice. In contrast, self-observation-based practice, in which performers independently diagnose and correct errors by observing their own movements through a mirror or video, can serve as a practical option[12]. However, these approaches largely depend on the performer's own interpretation and do not explicitly identify movement errors or indicate how they should be corrected. This limitation highlights the need for objective swing analysis methods that can quantify movement characteristics and support consistent feedback during self-practice.

Recent advances in sensing and deep learning technologies have broadened the range of golf swing analyses available to golfers. For example, commercial systems such as TrackMan combine radar and high-speed camera sensing to quantify ball and club parameters, including speed, spin, and carry, immediately after each shot[13]. Smartphone applications such as V1 Golf provide a more accessible option by enabling golfers to analyze their swings from RGB video alone, using features such as visual annotation and automatic ball and skeletal tracing[14]. In parallel, recent research has also explored intelligent coaching systems based on motion-related sensors and 3D motion reconstruction, which can provide quantitative kinematic feedback on swing execution and correction strategies[15–19]. Beyond visual-cue feedback, more recent systems with Large Language Models (LLMs) have extended this line of work by translating sensor-derived movement information into verbal feedback that humans can readily interpret[15].

Table 1. Comparison of golf swing datasets.

Dataset	Year	#Part	#Swings	Skill groups	Public open	#Mod.	Motion tracking	Foot press.	IMU signals	Gaze	FPV video	Ext. video	Shot out.
Nesbit et al.[1]	2005	85	85	PR&AM		1	✓						
Verikas et al.[20]	2016	15	~75	PR&AM		2	✓						✓
GolfDB[21]	2019	-	1400	PR.	✓	1						✓	
Navarro et al.[22]	2022	55	~550	PR&AM		3		✓				✓	✓
Ingwersen et al.[23]	2023	284	2907	PR&AM		1						✓	
Dong et al.[24]	2023	5	60	PR&AM		1	✓						
Kim et al.[25]	2023	36	108	PR&AM		2	✓		✓				
Lee et al.[26]	2024	6	20	PR&AM		2	✓					✓	
SolePoser[27]	2024	8	40	AM.		3	✓	✓	✓				
Kim et al.[28]	2024	20	400	PR&AM		2	✓		✓				
CaddieSet[29]	2025	8	924	AM.	✓	3	✓					✓	✓
Ours	**2026**	**24**	**1557**	**PR.&AM.**	✓	**7**	✓	✓	✓	✓	✓	✓	✓

Note. PR. denotes the professional group, and AM. denotes the amateur group. Check marks indicate characteristics explicitly reported as included. Modalities are counted as physically or semantically distinct sensing streams. The #Mod. column reports the total number of modalities described in each study. The seven modalities of MultiSenseGolf are: (1) full-body motion tracking, (2) foot pressure, (3) head IMU signals, (4) eye gaze, (5) first-person view (FPV) video, (6) external front-view RGB-D video, and (7) shot outcome data.

In this context, multimodal sensing can support more comprehensive golf swing analysis because different modalities capture distinct information about swing execution and its outcomes[8,30]. It can also improve robustness by supplementing information that is unavailable or unreliable in one modality with complementary information from another modality[27,31,32]. For example, external video can provide spatial information of body joints when they are visible, but rapid motion and self-occlusion can obscure body parts such as the hands and arms. Inertial motion capture (IMC)-derived joint kinematics remain available under these conditions, while visual observations can in turn be used to constrain accumulated drift in inertial measurement units (IMUs)[33,34]. However, current golf datasets rarely provide such complementary streams together. As summarized in Table 1, some existing datasets focused on a single modality or a limited subset of movement and performance information because they were designed for specific analysis tasks[1,21,24,26,29]. Others contained a limited number of recorded swings because they used controlled collection protocols or included golf as one activity in a broader study[24,26,27]. Lastly, some datasets were collected primarily for study-specific analyses and are not publicly accessible in practice[20,22,23,25,28].

To address this gap, we constructed MultiSenseGolf, a multimodal sensor-based dataset for golf swing motion and shot performance analyses. The purpose of this dataset is to support investigations of the relationships between golf swing motion and shot outcomes across different skill levels. It contains 1,557 swings from 24 participants in Beginner, Intermediate, and Professional groups, combining full-body joint kinematic estimates, foot pressure recordings, first-person view (FPV) video, gaze and head-motion data, front-view RGB-D video, and shot outcomes in swing-level records. These complementary observations are linked with participant metadata to support comparisons across swings, participants, and skill levels.

The Technical Validation section characterizes the dataset's measurement and timing properties, including temporal correspondence between sensor streams. It also presents a shot outcome prediction benchmark that assesses predictive performance under five single-modality input conditions. Together, these analyses provide a basis for interpreting the sensor recordings and reproducible reference results for the development and evaluation of shot outcome prediction models.

Methods

Sensor Modalities and Data Acquisition

The sensor setup used to collect the MultiSenseGolf dataset is illustrated in Figure 1. The wearable setup (a) consisted of an IMC system, a glasses-form-factor wearable sensor, and custom bilateral insole pressure sensors. The IMC system provided full-body joint kinematic estimates, the glasses recorded FPV video, gaze coordinates, and raw head-mounted IMU signals, and the insoles captured bilateral plantar pressure streams. The environmental setup (b) additionally included a stereo depth camera positioned in front of the participant for front-view RGB-D recording and an optical launch monitor that reported shot outcome metrics, such as carry distance or clubhead speed. The unit, sampling frequency, and measurement or estimation error of each modality are summarized in Table 2.

Figure 2 describes the data recording pipeline structures for each sensor modality. We adopted different approaches

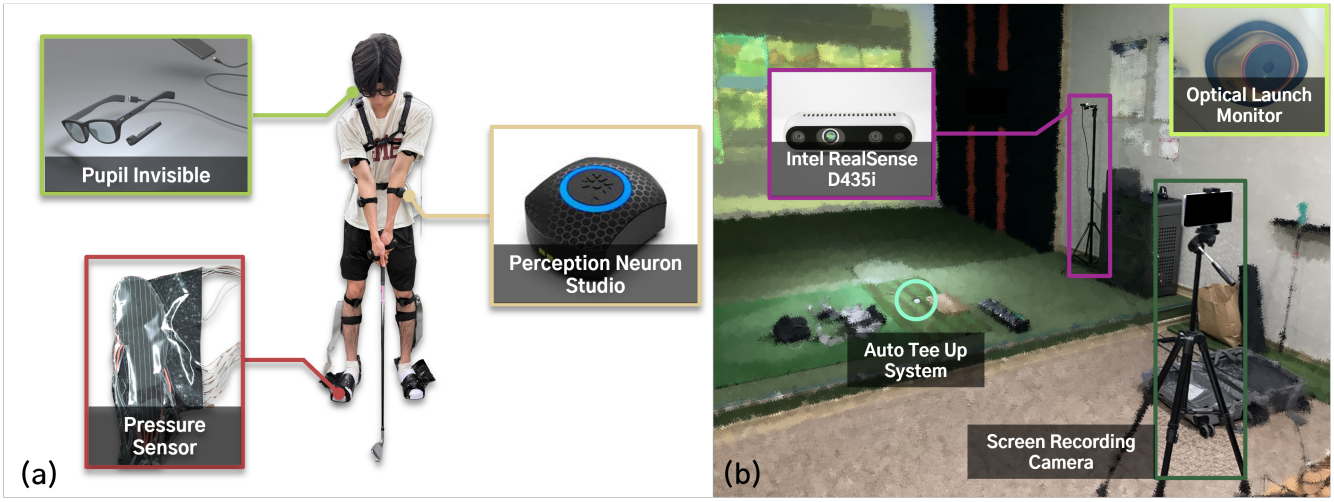

Figure 1. (a) Wearable sensors: Pupil Invisible (FPV video, gaze, head motion), Perception Neuron Studio (full-body IMC), and insole pressure sensors. (b) Environmental setup: Intel RealSense D435i (front-view RGB-D), optical launch monitor (shot metrics), screen-recording camera, and auto tee-up system.

depending on the data format and transmission characteristics. Numerical timeseries data, such as IMC-derived joint kinematic streams and pressure matrices, was collected using the ActionSense framework[38], a multimodal framework for recording data streams from heterogeneous sensors. In this work, we modified it to fit our environment, for example by adding a function to log the start and end timestamps of each swing[39]. In contrast, video streams were recorded separately using vendor-provided software tools to ensure stable capture. All recording pipelines were configured to store Unix time-based timestamps in their metadata. These timestamps were generated independently by each recording device (i.e., the host PC and the smartphone) using its native OS system clock, and no explicit clock drift correction was applied across devices at recording time.

Perception Neuron Studio (IMC-derived joint kinematic estimates)

The wearable 3D IMC system, Perception Neuron Studio (PNS)[35], was used to capture participants' swing motions as timeseries estimates of full-body joint positions and orientations. The hardware consisted of 17 IMUs, attached to the following body segments: *head, chest, upper back, lower back, hips, upper arms, forearms, hands, upper legs, lower legs, and feet (bilaterally where applicable)*. Each unit measured 43 mm × 33 mm × 20 mm, and was worn by fastening straps to the corresponding body segments. The gyroscope measurement range was ±2000 dps. The accelerometer range was set to ±30 g for sensors placed on the arms and legs, where high-acceleration movements were expected, and ±8 g for sensors on other body regions. Each sensor unit had a 280 mAh battery, which was sufficient to ensure continuous recording without battery depletion throughout the recording sessions.

Data packets acquired from the sensors were transmitted to Axis Studio, the vendor software for PNS supporting real-time 3D motion visualization and sensor configuration. The output frequency was set to 96 Hz for this dataset. The recorded IMC-derived joint kinematic streams included parent-relative positions and parent-relative orientations represented as unit quaternions and Euler angles for 21 joints. Here, parent-relative means that each non-root joint value is expressed in the coordinate frame of its parent joint in the kinematic hierarchy. Raw accelerometer and gyroscope measurements from individual units were not included. These streams were sent to the ActionSense framework through TCP/IP via a PnsStreamer, and stored in Hierarchical Data Format version 5 (HDF5) format. In addition, PnsStreamer computed per-joint angular velocity via quaternion-difference estimation and root-relative joint positions via FK, and stored both alongside the other signals.

To establish the alignment between each IMU and its corresponding body segment, a posture calibration procedure was also provided by Axis Studio. After all sensors were mounted, the participant sequentially held three predefined static postures, a T-pose, an S-pose, and an A-pose, with the golf club placed on the ground. Each posture was maintained for three seconds while the system recorded the sensor readings.

Pupil Invisible (FPV video, Gaze data, and Head IMU signals)

To collect golfers' head- and eye-related data, we employed Pupil Invisible, a lightweight glasses-form-factor wearable sensor. The mass of the glasses was 46.9 g, and they were connected to a smartphone via a wired cable for power supply and real-time data storage. To capture FPV video, we utilized the device's scene camera. The recording was performed in MP4 format with a resolution of 1088 px × 1080 px, and the field of view was 82° horizontally and 82° vertically. The frame rate was set to 20 fps

Table 2. Measurement variables of the MultiSenseGolf dataset, with name, unit, sampling frequency, and measurement or estimation error.

Name	Unit	Freq.	Measurement / estimation error
Perception Neuron Studio (PNS) - 3D inertial motion capture system, 17 IMUs			
Root- and parent-relative positions	cm	96 Hz	Joint position was obtained through forward kinematics (FK), so positional error could accumulate along the kinematic chain and become larger at joints farther from the root; if PNS orientation errors were similar to the vendor-specified attitude accuracy or task-based joint-angle Root Mean Squared Errors (RMSEs) reported in optical-reference validation studies[35,36], Root Sum Square (RSS) propagation indicated that the resulting positions were mathematically sensitive to shifts ranging from several millimeters to approximately 150 mm. These values characterized conditional FK sensitivity rather than measured positional errors in this dataset (see the Technical Validation section for details).
Euler orientation	degree (°)	96 Hz	Vendor-specified static attitude accuracy: 0.7° (roll/pitch) and 2.0° (yaw). Previous validation studies of PNS with optical-reference reported task-based joint-angle RMSEs of 1.9°-13.14° across joints, rotation axes, and tasks.[35,36]
Quaternion orientation	—	96 Hz	
Derived angular velocity	rad/s	96 Hz	Derived angular velocity was computed from finite differences between consecutive quaternions, inheriting orientation errors and amplifying finite-difference noise.
Derived acceleration	m/s^2	96 Hz	Derived acceleration was synthesized from joint positions via finite difference, so it inherited the positional error and amplified finite-difference noise.
Custom insole-type foot pressure sensor			
Left / right foot pressure	12-bit ADC	10 Hz	Pressure data was stored as raw 12-bit ADC values without per-sensor calibration to absolute units, supporting relative contact patterns and weight transfer analysis.
Left / right foot CoP	grid index	10 Hz	Center of Pressure (CoP) was derived as the weighted average of the pressure map.
Pupil Invisible — first-person view camera with onboard gaze and head-mounted IMU			
FPV video	8-bit RGB	20 fps	Occasional sample dropouts occurred when the wired connection to the recording device became unstable during dynamic swing motions.
Gaze	px	120 Hz	Reported mean angular error was approximately 5.5°, with a random spread of approximately 4° over most of the field of view[37]; same recording dropout issue as FPV video.
Head acceleration	m/s^2	200 Hz	Calibration-free; same recording dropout issue as FPV video.
Head angular velocity	deg/s	200 Hz	
Intel RealSense D435i — front-view RGB-D			
RGB video	8-bit RGB	30 fps	—
Raw depth image	m (0.001/unit)	30 fps	Vendor-reported Z-accuracy (absolute error) was within ±2% at ranges up to 2 m, with temporal noise below 1% (80% ROI, HD resolution).
Visualized depth image	8-bit RGB	30 fps	
Optical launch monitor — shot outcome metrics			
Total distance	m	per swing	
Carry distance	m	per swing	
Ball speed	m/s	per swing	
Club speed	m/s	per swing	
Launch angle	degree (°)	per swing	Measured by a commercial optical launch monitor and recorded as the device's reported values.
Horizontal launch angle	degree (°)	per swing	
Clubface angle	degree (°)	per swing	
Backspin	rpm	per swing	
Sidespin	rpm	per swing	
Lateral distance	m	per swing	

for this dataset, which allowed the FPV stream to provide egocentric visual context such as the spatial relationship between the player, the ball, and the ground. To support fine-grained head dynamics, the device's onboard IMU stream recording, including raw accelerometer and gyroscope (angular velocity) signals, was set to 200 Hz. In addition, we recorded 2D gaze points in the scene camera coordinate system at a sampling rate of 120 Hz. These were inferred by a neural network model from images captured by two infrared eye cameras embedded in the glasses frame.

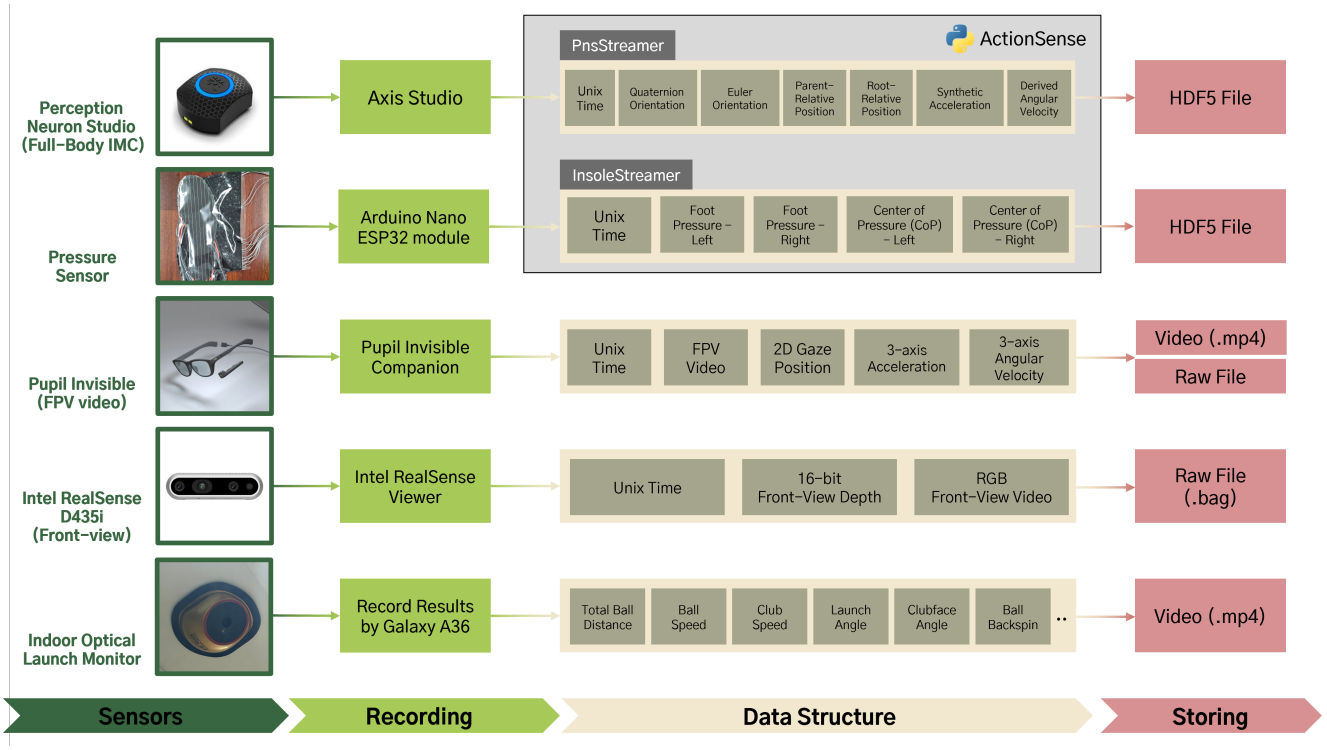

Figure 2. Data acquisition and storage pipeline. Full-body IMC and foot-pressure streams were recorded via the ActionSense framework[38] and stored in HDF5 format. FPV video was stored as an MP4 file, while gaze and head-mounted 6-DoF IMU streams (3-axis acceleration and 3-axis angular velocity) were stored as raw files. Front-view RGB-D data was recorded in BAG format, and shot outcomes were captured as MP4 screen recordings.

Custom-fabricated Insole-type Sensor (Plantar pressure data)

Foot pressure data serves as a fundamental source for extracting biomechanical features such as ground reaction force (GRF)[30,40] and center of pressure (CoP)[40]. Since this information provides physical context regarding ground contact and balance, it has been widely utilized in motion analysis domains ranging from gait stability[41,42] to sports analytics[43]. In the context of golf swing, kinetic factors such as GRF and momentum transfer have an intrinsic relation to impact power and carry distance[7,8]. Since pressure signals are robust to errors such as drift[40], we incorporated an insole-type foot pressure sensor into our data collection pipeline.

The insole pressure sensor used in this work was built based on a piezoresistive pressure-sensing array[44–47]. The sensor array consisted of electrode layers arranged orthogonally on the two sides of a piezoresistive film, where each electrode intersection functioned as an individual pressure sensor (see Figure 3). In this dataset, we used a 24×10 sensor grid per foot, yielding a total of 240 pressure measurement points. The sensing range of the device spanned from 0.3 kPa to 14 kPa for each individual sensing intersection[48]. To accommodate differences in participants' foot sizes, the insoles were fabricated in two lengths (240 mm and 270 mm). Both versions used the same pressure sensor array layout and the same number of sensing elements. As a result, the electrode spacing varied slightly with insole size, ranging from approximately 8 mm to 10 mm. This design ensured a proper fit across foot sizes while preserving a consistent spatial interpretation of the collected pressure data[46].

For communication and storage of the signals, the electrode layers of the insole sensor interfaced with an ESP32-based readout module through a pin-header connection. The module utilized a 12-bit analog-to-digital converter (ADC) to digitize the analog voltage changes induced by pressure. For power, we used a portable power bank (5,000 mAh) providing 5 V and 1 A output, and housed both the module and the battery in a detachable strap-mounted pouch worn on the calf. Pressure measurement data transmitted via ESP32 was received at a 10 Hz sampling rate through a receiver board connected to the host PC. It was then converted into frame-wise two-dimensional pressure maps by the InsoleStreamer module within the ActionSense framework and stored in HDF5 format as raw 12-bit ADC values. No per-sensor calibration to absolute pressure units (e.g., kPa) was conducted before data collection, as the dataset provided only raw ADC values to support analyses based on relative contact patterns and weight distributions[48].

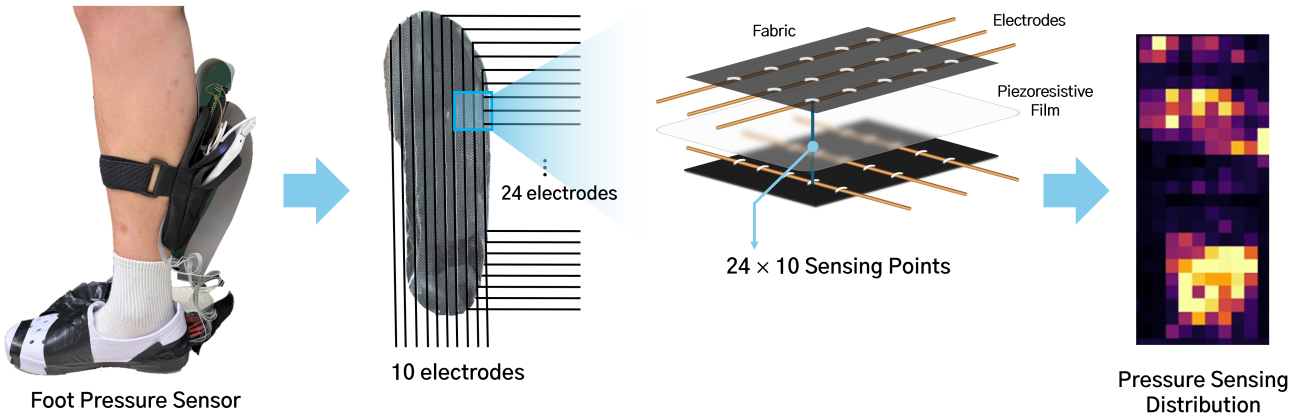

Figure 3. Custom insole-type foot pressure sensor. From left to right: in-shoe mounting, the 24 × 10 electrode layout, the multilayer sensing structure, and an example pressure heatmap.

Intel RealSense D435i (External videos)

To capture a frontal view of the participants throughout the swing, we installed an Intel RealSense D435i camera directly in front of the hitting area. RGB and depth streams were recorded at a resolution of 640 × 480 and a frame rate of 30 fps using Intel RealSense Viewer, vendor-provided software for stable data recording. During recording, we stored the raw data without post-processing or filtering. The recordings were saved in the RealSense native .bag format, which contained the raw RGB and depth streams with frame-level timestamps in a single file. The camera was operated using its factory calibration, and no additional user calibration was performed before data collection. The Intel RealSense D435i was designed for out-of-the-box operation and to maintain its calibration during use[49,50].

Shot Outcome Data

We collected shot measurement metrics as quantitative indicators of swing performance that reflected the resulting ball and club dynamics after each swing. The ceiling-mounted markerless optical launch monitor system captured ball and club motion at impact to compute key performance metrics, including ball speed, carry distance, launch angle, and club speed. These metrics were displayed as a result table on the front screen immediately after each shot. To digitize this data, we recorded the screen display continuously throughout the sessions. Using the swing timestamps, we identified the specific video frame containing the result table for each swing and manually transcribed the displayed values into the dataset.

Survey Data

To assess subjective discomfort induced by the wearable sensors, we administered the Comfort Rating Scale (CRS) questionnaire[51] after each data collection session. Participants evaluated each worn sensor, including PNS, Pupil Invisible, and the insole-type pressure sensor, using 11 items on a 7-point Likert scale covering six discomfort dimensions: emotion, attachment, harm, perceived change, movement, and anxiety. We also conducted a short interview in which participants ranked the sensors by perceived discomfort and explained their reasons. The interview responses were audio-recorded and included in the dataset as summarized transcripts.

Participants

A total of 24 participants were recruited for data collection (10 female, 14 male; mean age = 41.83, SD = 14.75). Participants were evenly divided into three skill groups: Beginner, Intermediate, and Professional (n = 8 per group). Eligibility for all groups required at least 3 months of golf experience. For this study, we classified non-professional golfers into two groups based on golf experience, using study-specific thresholds: Beginner if they had 3 months to 2 years of experience, or Intermediate if they had more than 2 years of experience. Inclusion in the Professional group required credentials issued by the Korea Professional Golfers' Association (KPGA), Korea Ladies Professional Golf Association (KLPGA), United States Golf Teachers Federation (USGTF), or Korea Teaching Professional Golfers' Association (KTPGA). Individuals whose physical condition prevented them from performing golf swings, for example because of an injury, were excluded from participation in the study.

All participants performed right-handed golf swings, with the left side serving as the lead side, while their self-reported dominant hand was distributed as follows: 20 right-handed and 4 left-handed. Detailed participant-level information, including golf experience, professional license, and anthropometric measures, is provided in Table 3 and the dataset.

Table 3. Participant demographics and golfing experience.

ID	Gender	Age	Swing Side	Dominant Hand	Skill Level	Golf Experience	Pro License
P01	Female	23	Right	Right	Beginner	3 to 6 months	-
P02	Female	47	Right	Right	Beginner	6 months to 1 year	-
P03	Female	55	Right	Right	Beginner	1 to 3 years	-
P04	Male	20	Right	Right	Beginner	3 to 6 months	-
P05	Female	58	Right	Right	Beginner	3 to 6 months	-
P06	Female	57	Right	Right	Beginner	6 months to 1 year	-
P07	Male	22	Right	Right	Beginner	3 to 6 months	-
P08	Male	35	Right	Right	Beginner	6 months to 1 year	-
P09	Male	52	Right	Right	Intermediate	3 to 5 years	-
P10	Male	38	Right	Right	Intermediate	3 to 5 years	-
P11	Female	58	Right	Left	Intermediate	More than 10 years	-
P12	Male	52	Right	Right	Intermediate	More than 10 years	-
P13	Female	52	Right	Left	Intermediate	3 to 5 years	-
P14	Male	30	Right	Right	Intermediate	1 to 3 years	-
P15	Male	57	Right	Right	Intermediate	More than 10 years	-
P16	Male	20	Right	Right	Intermediate	3 to 5 years	-
P17	Male	64	Right	Left	Professional	More than 10 years	USGTF Teaching Professional
P18	Male	58	Right	Left	Professional	More than 10 years	KTPGA Teaching Professional
P19	Male	51	Right	Right	Professional	More than 10 years	KPGA Associate Member
P20	Male	31	Right	Right	Professional	More than 10 years	KPGA Associate Member, USGTF Teaching Professional
P21	Female	20	Right	Right	Professional	3 to 5 years	USGTF Teaching Professional
P22	Male	31	Right	Right	Professional	More than 10 years	USGTF Teaching Professional
P23	Female	38	Right	Right	Professional	More than 10 years	KLPGA Associate Member
P24	Female	35	Right	Right	Professional	More than 10 years	USGTF Teaching Professional

Note. *Swing Side* indicates the direction of the golf swing, while *Dominant Hand* indicates the participant's self-reported primary hand for general use.

Data Collection Setup

Data collection was conducted in an indoor golf practice facility to maintain consistent environmental conditions across participants and sessions. This setting provided a private space, which helped limit concerns related to being observed by others while participants performed swings with wearable equipment. In addition, for video-based data, outdoor recording could lead to fluctuations in image quality and degraded depth estimation due to changes in weather and illumination. The indoor facility also enabled reliable acquisition of shot outcome measurements, such as launch angle and ball speed, which are often difficult to obtain consistently in outdoor settings.

Regarding the experimental equipment, all participants used a single, standard-size 7-iron club provided by the experimenters. We selected the 7-iron as a representative club type because it is widely used as a standard reference suitable for different skill levels[52]. This choice also helped reduce invalid trials among novice participants, for whom other club types such as drivers might be more difficult to handle. Moreover, including multiple club types would have required a larger number of swings, increasing the risk of physical fatigue and degradation in motion quality.

Figure 1 (b) illustrates the experimental environment and the placement of sensors and recording devices. The facility was equipped with a built-in auto tee-up system that automatically supplied a new ball for each swing. An optical launch monitor was installed on the ceiling above the hitting area to track ball and club dynamics. For video recording, an Intel RealSense D435i was positioned to the right of the screen to capture the front view. Simultaneously, a Galaxy A36 smartphone was mounted on the opposite side to record the participant's side view along with the shot measurement data displayed on the screen.

Procedure

Before the experiment, participants received an overview of the procedure and completed a questionnaire on basic demographics and background information, including height, body mass, and golf experience. Participants then wore PNS, Pupil Invisible, and the insole-type pressure sensor on the designated body locations with the assistance of one or two experimenters. After participants performed practice swings for acclimation and warm-up, we calibrated the PNS IMC system following the aforementioned procedure. This procedure was repeated until the experimenters confirmed stable calibration.

In each session, participants performed repeated swings using a provided 7-iron club. To account for potential performance changes and data quality degradation due to fatigue, we set different minimum target numbers of swings by skill group. Participants in the Beginner and Intermediate groups were asked to complete at least 50 swings, whereas participants in the Professional group were asked to complete at least 100 swings. They were also instructed to proceed at their typical pace and to take breaks as needed. The experimenters provided progress updates in increments of 10 swings, checked the recording status, and re-ran PNS calibration when they deemed it necessary. After participants completed the target number of swings, sensors were removed and the session concluded with the questionnaire and interview. Session duration was approximately 1.5–2 hours for the Beginner and Intermediate groups and approximately 2–2.5 hours for the Professional group. Compensation was tiered by skill group: Beginner, 40 USD; Intermediate, 50 USD; and Professional, 100 USD.

Ethics Statement

The study involving human participants was approved by the Institutional Review Board of the Gwangju Institute of Science and Technology (IRB approval number: HR-84-08-04). Participants were recruited through online platforms and were provided with information about the study before participation. All methods were carried out in accordance with relevant guidelines and regulations. Prior to participation, all participants were fully informed about the purpose, procedure, and types of data to be collected. They provided written informed consent for their data to be publicly shared for research purposes, including the publication of anonymized images in the figures. To protect participant privacy, all identifiable facial features in the videos and images were anonymized using blurring techniques, and all personally identifiable information (PII), such as names, was excluded from the released dataset.

Data Processing

The raw session-level recordings included not only valid swing trials, but also periods of sensor calibration, participant rest, recording-status checks, and unsuccessful swing attempts. These full-session recordings resulted in a substantial storage burden of approximately 2.56 TB, primarily due to the front-view and FPV video files. We therefore implemented a systematic data processing pipeline that converted the original session-level recordings into swing-level segments. For the front-view recordings, RGB frames, depth images, and frame-level timestamps were additionally extracted from the uncompressed raw files. For each modality, only the frames or samples whose timestamps fell within the annotated swing interval were retained. The released dataset now occupies approximately 45 GB, representing a reduction of about 98%. This process also enabled us to eliminate the need for users to perform repetitive segmentation and alignment, allowing for immediate use in per-swing comparison or machine learning tasks.

Figure 4 illustrates the overview of the full data processing pipeline consisting of three sequential stages. First, in *Swing Timestamp Refinement*, we manually refined the approximate real-time logs into precise swing boundaries. Next, *Impact-based Timestamp Offset Correction* was performed by manually annotating timestamps of the nearest frame with the impact event from front-view videos and FPV videos and correcting temporal offsets across them. Finally, in the *Swing-Based Data Slicing* stage, these refined boundaries were used to segment the raw multimodal streams into individual swing samples.

Swing Timestamp Refinement

To extract individual swings from the raw recordings, it was necessary to define the start and end times of each swing in the common time unit. Therefore, the experimenter observed the participant's movements and used the GUI provided by the ActionSense framework[38] to mark swing start and end timestamps in the form of Unix Time during data collection. However, as these raw timestamps indicated only the approximate temporal location of each swing, we performed an additional post-hoc refinement procedure. Using the ELAN video annotation tool[53], we reviewed the front-view MP4 video and manually refined the start and end boundaries for each swing. Specifically, we defined the swing start as the timestamp of the first frame in which the participant assumed the address posture and became stationary, and the swing end as the timestamp of the last frame immediately before the participant exited the finish posture. No additional pre- or post-buffer time was added beyond these swing boundaries.

Impact-based Timestamp Offset Correction

Unix timestamps could still differ between independently clocked recording systems due to clock offset and drift[54], as well as differences in data acquisition and processing latency[55]. We therefore applied an event-based timestamp offset correction

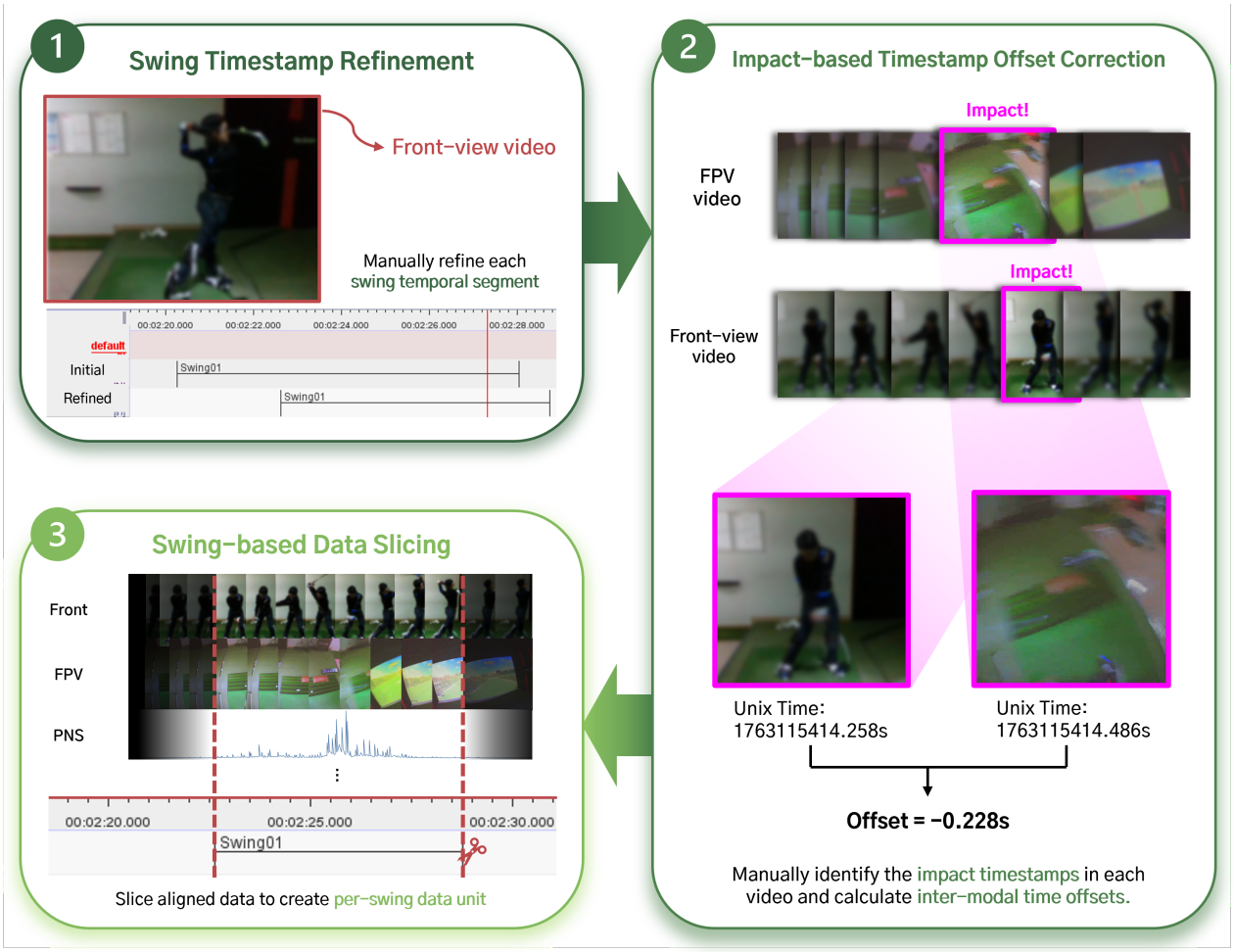

Figure 4. Post-hoc Data Processing Pipeline. (1) Manual refinement of per-swing temporal boundaries using front-view video, (2) Impact-based timestamp offset correction between FPV and front-view videos to compute the inter-modal time offset, and (3) Slicing of all sensor streams into per-swing data units using the refined boundaries and offsets.

procedure using an *impact* as a common reference event, defined as the moment of club-ball contact[21]. To estimate the observed offset between the two recording systems, we selected the FPV video recorded by the smartphone and the front-view RGB video recorded by the host-PC-based system as representative streams. For each stream, we manually annotated the timestamp of the frame nearest to the impact moment and calculated the offset as the difference between the two timestamps. Because the exact physical instant of contact could not be resolved at the finite frame rates of the video streams, these annotations were subject to a timing uncertainty[56]. We further characterized this uncertainty in the Technical Validation section.

Using the observed offset for each swing, we adjusted the FPV frame timestamps to the front-view time axis. Because the gaze and head-mounted IMU streams were recorded through the same smartphone-based recording system and referenced the same hardware clock as the FPV stream, the same offset was applied to their sample timestamps. This procedure corrected the observed timestamp difference for swing-level slicing but did not independently establish the synchronization accuracy of the streams. For reproducibility, the dataset provides the manually annotated impact frame index and corresponding Unix timestamp for both the FPV and front-view videos.

Data Slicing

Finally, we segmented the session-level sensor streams into swing-level records using the refined start and end timestamps obtained in the previous steps. For each swing, we retained the frames whose timestamps fell within the inclusive interval defined by these boundaries. In particular, the original timestamps were used for the front-view, PNS, and insole pressure streams, whereas the offset-adjusted timestamps described in the previous step were used for the FPV, gaze, and head-mounted IMU streams. Shot outcome data was not included in these procedures because it was recorded once per swing rather than as a timeseries. The detailed computational procedure for the three stages of swing timestamp refinement, timestamp offset

Algorithm 1: Swing-level timestamp offset correction and data slicing

Input : modality streams $\mathscr{D}^{(m)}$, front-view and FPV videos
Output : swing-level records $X_i^{(m)}$

for *each swing i* **do**
  Identify the first stationary address frame and define its timestamp as s_i
  Identify the last frame before exiting the finish posture and define its timestamp as e_i
  Annotate the FPV and front-view frames nearest to impact as t_i^{FPV} and t_i^{FV}
  Compute the observed timestamp offset:

$$\Delta_i = t_i^{\mathrm{FV}} - t_i^{\mathrm{FPV}}$$

for *each swing i and modality m* **do**
  for *each timestamp index k* **do**
    Define the timestamp used for slicing:

$$\tilde{t}_{i,k}^{(m)} = \begin{cases} t_{i,k}^{(m)}, & m \in \{\mathrm{FrontView}, \mathrm{PNS}, \mathrm{Pressure}\}, \\ t_{i,k}^{(m)} + \Delta_i, & m \in \{\mathrm{FPV}, \mathrm{Gaze}, \mathrm{HeadIMU}\}. \end{cases}$$

  Define the swing-level record:

$$X_i^{(m)} = \left\{ x_{i,k}^{(m)} \;\middle|\; s_i \le \tilde{t}_{i,k}^{(m)} \le e_i \right\}$$

correction, and data slicing is provided in Algorithm 1.

During this process, we performed several computations as post-processing steps. To generate visualized depth frames, the *pyrealsense2* post-processing filter pipeline was applied, which consisted of downsampling and valid-range clipping, followed by sequential spatial and temporal smoothing in disparity space. Moreover, CoP was computed as a weighted average normalized by the total pressure at each time point from the foot pressure streams.

Data Records

The MultiSenseGolf dataset, providing swing-level data records that integrate multimodal wearable sensor streams with video-based observations, is publicly available via Harvard Dataverse[57]. The dataset includes a total of 1,557 multimodal golf swings, with each swing record organized in a directory structure indexed by participant and swing IDs. For each swing, we provide video files, numerical sensor streams, and timestamp metadata at the frame and sample levels. The timeseries visualization of a representative single swing sample is illustrated in Figure 5.

The final dataset contains 401 Beginner swings, 323 Intermediate swings, and 833 Professional swings. These totals differ from the expected totals based on the minimum acquisition targets because final inclusion was determined through post hoc review and data quality screening. Regarding exclusions, recorded attempts were omitted when the front-view recording was unavailable because of a missed recording or streaming failure. Because the front-view video served as the reference for swing boundary refinement, valid swing records could not be generated from these attempts. Recorded attempts were also excluded when streaming failures made data from more than half of the recorded modalities unavailable. These exclusions primarily accounted for the Intermediate group containing fewer than the expected 400 swings, particularly because of front-view recording failures during several sessions such as *P12* and *P13*. In contrast, some participants performed a small number of additional attempts to compensate for anticipated recording loss. During post hoc review of the complete front-view recordings, we also identified valid swings that were absent from the real-time start and end timestamp logs. These swings were subsequently annotated and included in the released dataset. Participants in the Professional group generally performed slightly more than the minimum target of 100 swings, contributing to the final total of 833 valid swings.

File Structure

Figure 6 summarizes the file and directory organization of the released dataset. The dataset is categorized into two parts, *Data* and *Documentation*, according to the type of content provided. *Data* contains the curated multimodal sensor streams and is

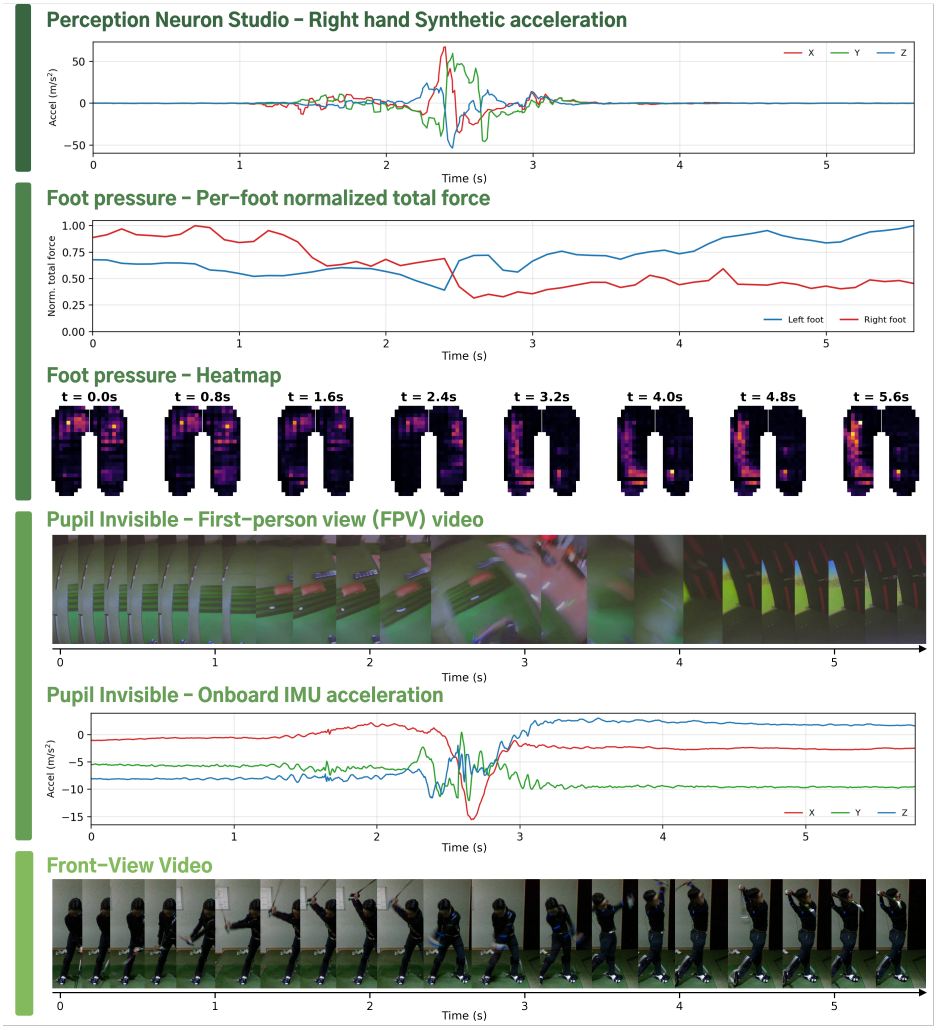

Figure 5. Representative multimodal recording of a single golf swing (P24, Swing05). From top to bottom: PNS right-hand synthetic acceleration; per-foot normalized total force for both feet; a temporal sequence of foot-pressure heatmaps; FPV video frames; Pupil Invisible IMU acceleration; and front-view RGB frames. Video frames were sampled at evenly spaced timestamps between the first and last frames.

provided as individual compressed archives (ZIP) for each participant to accommodate storage constraints. For the professional group (*P17–P24*), which contains a larger number of swings, the data for each participant is further split into multiple archives. Specifically, for *P17* through *P23*, each participant's data is divided into two parts, while data for *P24* is divided into three parts. Within each participant directory, data is organized into swing-indexed subdirectories (e.g., Swing01). Each swing directory stores (i) front-view data recorded with an Intel RealSense D435i, (ii) an FPV video file, and (iii) an HDF5 file containing the remaining numerical sensor streams. *Documentation* includes participant metadata, annotation data including shot measurement metrics, and summarized survey and interview materials.

Description of *Data*

HDF5 Files

Each swing directory contains an HDF5 file that consolidates numerical sensor streams, following the filename convention *{participant_id}_{swing_id}_stream_data.hdf5* (e.g., P01_Swing01_stream_data.hdf5). The HDF5 structure is organized as a top-level group per modality. Each modality group contains stream-level subgroups and a shared set of keys: *data*, *time_s*, and *time_str*. The *data* field stores observed values, *time_s* stores Unix time-based timestamps for each sample as seconds in floating-point format, and *time_str* stores the same timestamps as human-readable datetime strings. This structure is applied consistently across the numerical streams such as insole pressure, PNS motion tracking, and Pupil Invisible gaze and IMU streams. The full hierarchy of groups, streams, and their array shapes for each modality is illustrated in Figure 6 (a).

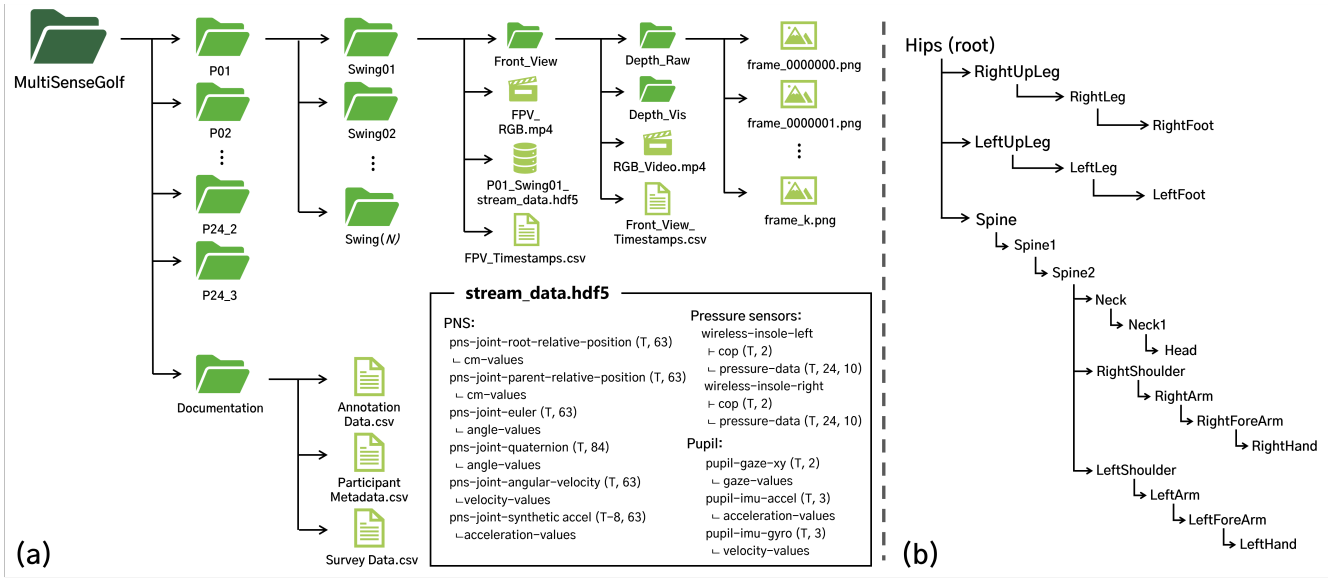

Figure 6. (a) Hierarchical folder structure of the dataset. Data and Documentation contain per-participant sensor recordings and supporting documents, respectively. A box below shows the internal group structure of each per-swing HDF5 file with array shapes per modality. T denotes the number of time steps per swing, and all groups share data, time_s, and time_str keys for values and timestamps. (b) Joint hierarchy of the PNS skeleton. The *Hips* joint serves as the root, and the branches indicate which joint serves as the parent of each non-root joint within the 21 joints used to represent positions and orientations.

The IMC outputs are stored as separate streams for two types of joint positions (i.e., parent-relative, root-relative), two representation types of parent-relative orientations (i.e., Euler orientations, quaternion orientations), derived angular velocities, and synthetic acceleration. Each stream contains values for 21 joints, although the synthetic acceleration stream is shorter by eight samples because of the finite-difference window (N=4). The joint order used in this dataset is as follows: *Hips, RightUpLeg, RightLeg, RightFoot, LeftUpLeg, LeftLeg, LeftFoot, Spine, Spine1, Spine2, Neck, Neck1, Head, RightShoulder, RightArm, RightForeArm, RightHand, LeftShoulder, LeftArm, LeftForeArm, and LeftHand.* Among these joints, the *Hips* joint serves as the root of the joint hierarchy. Figure 6 (b) illustrates the complete joint hierarchy of the PNS skeleton, including the parent-child relationship of each non-root joint. This hierarchy serves as a reference for interpreting the parent-relative positions and orientations by identifying the parent joint frame associated with each non-root joint. To enable users to apply the processing most appropriate to their specific application, all motion capture streams are provided as exported by the capture system, without any preprocessing other than the timestamp offset correction described above.

The coordinate and rotation conventions used for these streams are as follows. Axis Studio uses a right-handed coordinate system in which the positive X-axis points toward the participant's left, the positive Y-axis points upward, and the positive Z-axis is aligned with the participant's forward direction in the calibration pose. For each non-root joint, the parent-relative position represents its translation with respect to its parent joint, while the Euler and quaternion orientations represent its rotation with respect to the parent joint's coordinate frame. For the *Hips* root joint, the corresponding position and orientation are expressed with respect to the coordinate frame configured in Axis Studio. The dataset does not provide any absolute positions referenced to an independently calibrated laboratory coordinate system. Euler orientations are stored as three-component angle triplets and follow an intrinsic Y-X-Z rotation sequence. Quaternion orientations are stored as unit quaternions in (W, X, Y, Z) order, where W is the scalar (real) component. The derived angular velocities and synthetic accelerations follow the same joint order as the position and orientation streams. Angular velocities were computed post hoc from consecutive unit quaternions, whereas accelerations were synthesized from the joint positions using finite differences. Therefore, neither stream represents raw measurements from the body-worn IMUs. This distinction applies only to the PNS streams; the dataset separately provides acceleration and angular-velocity measurements recorded directly by the Pupil Invisible head-mounted IMU.

Insole pressure data is stored in separate groups for the left and right feet (e.g., *wireless-insole-left/pressure-data*). For each foot, the pressure data provides foot pressure distributions at each time step in the insole sensor grid coordinate system. In this grid, row index 0 corresponds to the top of the pressure map and row index 23 to the bottom, while column index 0 corresponds to the left side and column index 9 to the right side. We additionally provide a CoP stream for each foot, whose x and y coordinates are defined with respect to the column and row indices of the pressure map, respectively.

The Pupil Invisible gaze stream (e.g., *pupil-gaze-xy/gaze-values*) contains 2D gaze coordinates in the FPV video frame

coordinate system, where the origin is at the top-left corner of the frame, the x-axis increases rightward, and the y-axis increases downward. The accompanying IMU streams (e.g., *pupil-imu-accel/acceleration-values*) provide 3-axis acceleration and 3-axis angular velocity. Although the acceleration stream was originally recorded in units of g ($1g = 9.80665\,\mathrm{m/s^2}$), its values were converted to $\mathrm{m/s^2}$ in the released files to maintain consistent units across the acceleration streams, including the synthesized joint accelerations derived from the IMC estimates. From the wearer's first-person perspective, the x-axis points to the right side of the glasses, the y-axis points downward, and the z-axis points in the forward viewing direction.

Video Data

FPV video is provided as MP4 files containing RGB frames sliced to each swing segment. For all swings, the filename is standardized as *FPV_RGB.mp4*, with a resolution of 1088 × 1080 and a constant frame rate of 20 fps. FPV video is stored within each swing directory to enable swing-level access alongside the other modalities. Each swing directory also includes a timestamp CSV that records per-frame timing information for the FPV stream, which contains the fields *frame_index_in_swing*, *timestamp_unix_s*, and *timestamp_str*.

Front-view data consists of RGB and depth streams, stored under the *Front_View* folder within each swing directory. Depth is provided in two versions. For quantitative analysis, *Front_View/Depth_Raw* provides the sensor's original depth units as 16-bit single-channel PNG frames. The depth scale is set to 0.001 m per unit, such that a raw value of 1000 corresponds to 1 m. For rapid visual inspection, *Front_View/Depth_Vis* provides separate visualization depth frames using a Jet-type colormap ranging from approximately 0.3 m to 5 m. RGB is provided as swing-level MP4 videos. To protect participant privacy, a Gaussian blur filter was applied to faces in all the RGB video streams. We also provide a timestamp CSV that records within-swing frame indices and Unix time timestamps for RGB and depth frames in both seconds and datetime string formats.

Description of *Documentation*

The *Documentation* directory provides supporting materials for dataset interpretation and reproducible analysis. Because the numerical and video data is released as swing-level segments, we include document-based metadata that enables linking these records to participant characteristics, experimental conditions, and per-swing outcome metrics. Documentation includes an annotation table summarizing swing segments and shot outcomes, participant metadata including demographics and golf experience, and survey data capturing sensor wearing experiences and open-ended feedback. All files are provided in CSV format to improve machine readability. We expect that these documents will help users understand the data structure, verify variable definitions and units, and conduct participant-level and swing-level downstream analyses.

Annotation Data.csv

Annotation Data.csv provides an annotation table that integrates swing segment information, impact events, and shot outcome metrics reported by the indoor golf facility. Each row corresponds to a single swing and includes participant IDs, swing indices, and swing start and end times in Unix time seconds. Impact events are recorded for both FPV and front-view streams, including the within-swing frame index and the Unix time timestamp in seconds. Shot outcome metrics include ball and club distances, speeds, angles, and spin values. Distances are provided in meters, speeds in m/s, angles in degrees, and spin in rpm. Lateral distance values include an L or R prefix to indicate left or right direction.

Participant Metadata.csv

Participant Metadata.csv provides participant-level metadata, including demographics and golf experience. Each row corresponds to one participant and includes age, gender, height (cm), body mass (kg), swing handedness, hand dominance, and shoe size (mm). Golf experience and lesson experience are recorded as categorical variables with the following bins: No prior experience, 1 to 6 months, 6 months to 1 year, 1 to 3 years, 3 to 5 years, 5 to 10 years, and More than 10 years. Whether a participant holds a professional credential is provided as a binary value, and the credential type is recorded as a string field allowing multiple entries. Self-reported skill level was collected on a 7-point scale, where 1 indicates novice and 7 indicates expert. Data sharing consent indicates whether the participant consented to release their full data.

Survey Data.csv

Survey Data.csv provides questionnaire results on sensor wearing experiences by participant and by sensor. Each row corresponds to a participant–sensor pair and includes quantitative ratings and open-ended feedback. Quantitative ratings include the CRS subscale scores: Emotion, Attachment, Harm, Perceived Change, Movement, and Anxiety, where 1 indicates strongly disagree and 7 indicates strongly agree. Because some scores were computed as the mean of two items assessing the same dimension, values may include 0.5 increments. Sensor labels include PNS, insole-type pressure sensor, and Pupil Invisible. Open-ended discomfort feedback is provided as English translations and summaries of the original Korean responses.

Technical Validation

The Technical Validation section presents four analyses that characterize the MultiSenseGolf dataset from different perspectives. First, we quantified the mathematical sensitivity of PNS-derived joint positions to orientation errors propagated through forward kinematics (FK), assuming error magnitudes similar to the vendor-specified attitude accuracy and those reported in previous optical-reference validation studies of PNS. Second, we statistically analyzed the carry distance distribution across skill level, gender, age group, golfing experience, and hand dominance. Third, we characterized the observed cross-device timestamp offsets used for impact-based correction, their within-session stability, and the temporal correspondence between the hand acceleration peak and observed club-ball contact. Finally, we conducted a compact shot performance prediction benchmark grounded in the shot outcome data.

PNS Joint Position Sensitivity to Assumed Orientation Errors

To quantify how orientation uncertainty can propagate to the PNS-derived joint positions, which is one of the main sources of joint positional error in IMC systems, we conducted an FK-based positional sensitivity analysis using the dataset's joint hierarchy and the inter-joint lever arms observed across all 1,557 swings (580,744 frames). Two sets of orientation error inputs were evaluated: the vendor-specified attitude accuracy under static conditions (roll/pitch: 0.7°; yaw: 2.0°) and task-based, axis-specific joint-angle Root Mean Squared Errors (RMSEs) reported in optical-reference (i.e., OptiTrack) validation studies of PNS (1.9°-13.14° across the assessed joints, rotation axes, and movement tasks)[35,36]. For each upstream joint and rotation axis, the positional displacement of a child joint was calculated using the perpendicular distance from the rotation axis, $r_{\perp}$, as $d = 2r_{\perp}\sin(\theta/2)$. The displacement contributions induced by orientation errors at all parent joints were then combined using the Root Sum Square (RSS) for each joint. If the PNS orientation errors were similar to the vendor-specified attitude accuracy or the errors reported in the previous validation studies[35,36], the joint positions obtained by FK were mathematically sensitive to shifts ranging from several millimeters to approximately 150 mm, depending on the joint, rotation axis, and assumed error input.

Even though these results quantified the sensitivity of the reconstructed joint positions to the orientation errors, they should not be interpreted as estimates or bounds of PNS positional accuracy because this analysis did not constitute a comparison against independently measured joint position ground truth. Depending on the modeling assumptions, the resulting values might either overestimate or underestimate the actual positional errors. For example, the analysis assumed that the orientation error at each upstream joint contributed independently to the displacement of a descendant joint. This assumption might overestimate the displacement if adjacent joints shared a common orientation error, because a rotation affecting both the parent and child similarly could be reduced when the child orientation was expressed relative to the parent. In contrast, additional error sources such as sensor location drift, calibration error, and magnetic disturbance were not explicitly modeled, which might lead to underestimation.

Statistical Analysis by Participant Characteristics

To help users understand how the dataset is distributed across participant attributes, we conducted a statistical analysis of carry distance across five participant characteristics: skill level, gender, age group, golf experience, and hand dominance. Carry distance is a widely used shot outcome metric for examining the effects of individual characteristics[58] and biomechanical factors[59] on golf performance. We have also released the tutorial analysis script (see Code Availability) to support users in applying the same analysis to other metrics or participant characteristics.

To ensure independence across data points, each participant contributed a single representative carry distance, computed as the mean of their swings after removing per-participant outliers using the 1.5 × IQR rule. After assessing normality within each group via the Shapiro-Wilk test, we applied a one-way ANOVA (skill level, age group, golf experience), a Welch's t-test (hand dominance), or a Mann-Whitney U test (gender), with Bonferroni-corrected post-hoc pairwise comparisons when ANOVA indicated a significant difference ($p < .05$).

Figure 7 summarizes the carry distance distribution across the five characteristics. Skill level ($F(2, 21) = 20.15$, $p < .001$) and golf experience ($F(2, 21) = 16.12$, $p < .001$) both showed significant effects, with mean carry distances in the Beginner group (M = 57.90, SD = 27.13) and the < 1 year group (M = 55.52, SD = 28.40) significantly smaller than those in their respective comparison groups (all $p < .01$). This pattern suggests that carry distance differences in the current sample are primarily associated with the transition from early-stage to established practice. Male participants (M = 114.28) had significantly longer carry distances than female participants (M = 74.18). For hand dominance, four left-dominant participants performed right-sided swings during data collection. Although this cross-dominant swing style can affect shot outcome metrics[60], no significant difference was observed between the right-dominant (M = 96.81) and left-dominant (M = 101.38) groups, likely due to the imbalance in group sizes. These results should be interpreted with caution because several characteristics were not fully independent in the current sample. Users planning sensitive analyses should account for these confounds through stratified sampling or covariate control.

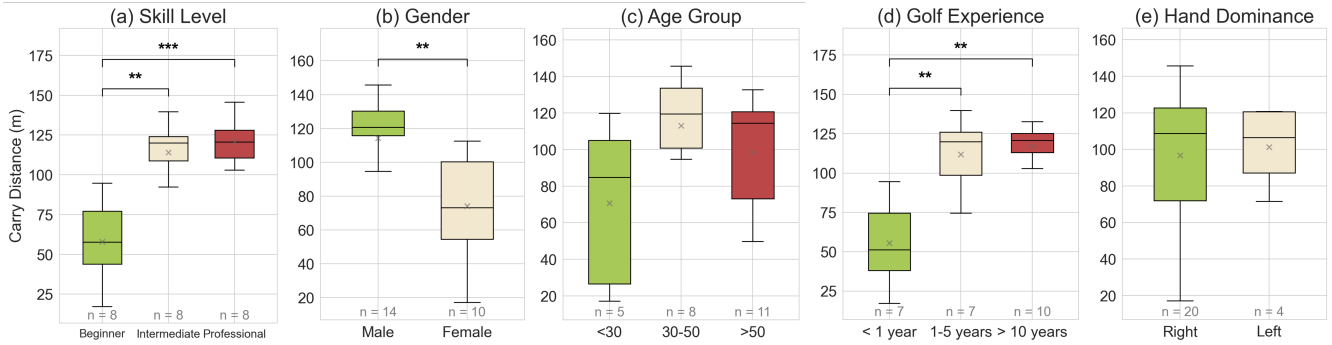

Figure 7. Statistical analysis results of the carry distance distribution across the five characteristics: (a) Skill level, (b) Gender, (c) Age group, (d) Golf experience, (e) Hand dominance. Significance levels are indicated: ** $p < .01$, *** $p < .001$

Temporal Offset and Variability Characterization

Characterizing the Timestamp Offsets Used for Correction

To characterize the observed timestamp offsets used in the correction procedure described in the Data Processing section, we conducted three complementary analyses. First, we examined the magnitude and distribution of the per-swing timestamp offsets derived from the visually annotated impact events in the FPV and front-view videos. For each swing, we computed the observed offset as the front-view impact timestamp (host PC clock) minus the FPV impact timestamp (smartphone clock) and aggregated the resulting offsets across all 1,184 swings with both impact annotations available. Because the same impact annotations were used to derive the correction values, this analysis does not provide an independent assessment of post-correction synchronization accuracy. As shown in Figure 8 (a), the per-swing offset distribution showed a mean offset of 699.85 ms (median 870 ms, SD 665.65 ms, range -540 to 2410 ms), demonstrating substantial variation in the pre-correction timestamp differences across swings. Note that these values describe the timestamp differences observed before correction. That is, the released swing-level streams were generated after applying the corresponding per-swing offset corrections.

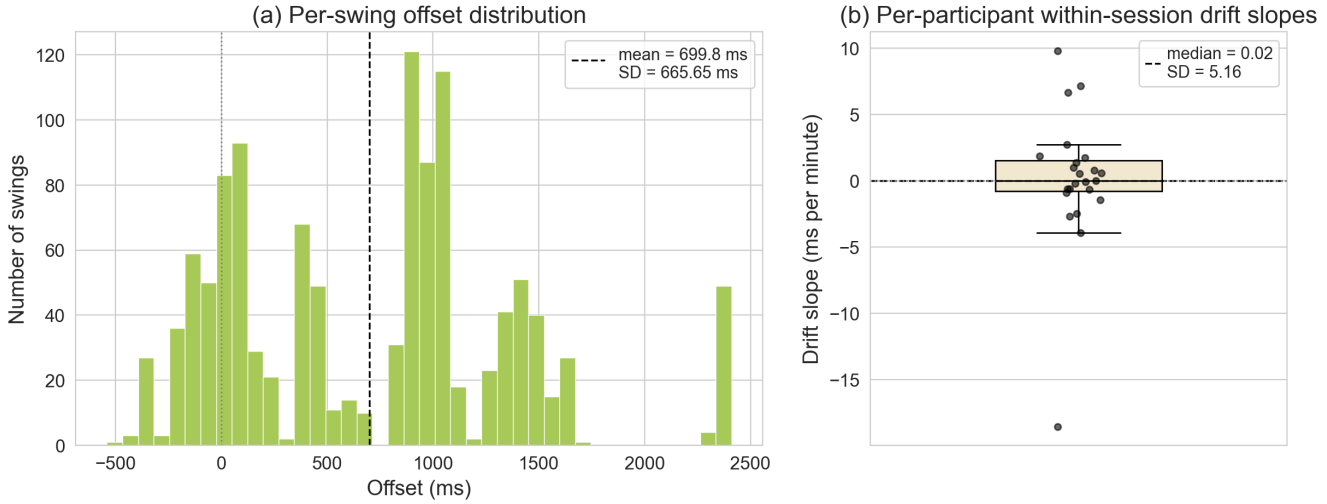

Figure 8. Timestamp offsets used for correction and their within-session trends. (a) Distribution of per-swing pre-correction offsets, calculated by subtracting the FPV impact timestamp from the front-view impact timestamp. A dashed line indicates the mean offset. (b) Participant-specific within-session slopes obtained by regressing the observed offset against elapsed session time.

Next, we quantified within-session changes in the observed offsets across participants. For each participant, we used the same per-swing offsets as defined above and treated the front-view impact timestamp of the first swing in the session as the reference point. The offset of each swing was then plotted against its elapsed time from this reference, and a linear regression was fitted to obtain a participant-specific within-session offset slope in milliseconds per minute[54,61]. This regression was performed for each participant with at least three valid swings. As shown in Figure 8 (b), the resulting slopes were centered

close to zero (median 0.018 ms/min, mean 0.09 ms/min), suggesting that a progressive change in the observed offsets was not consistently present across participants. However, the slope estimates showed substantial between-participant dispersion, with a sample SD of 5.16 ms/min. Four participants (*P04*, *P10*, *P11*, and *P15*) had slope estimates outside Tukey's fences[62], defined as values below $Q_1 - 1.5 \times \text{IQR}$ or above $Q_3 + 1.5 \times \text{IQR}$. To examine whether the sample SD was sensitive to the extreme slope estimates, we additionally calculated a robust SD based on the median absolute deviation (MAD)[63], defined as $1.4826 \times \text{MAD}$. The resulting robust SD was 1.98 ms/min, which was smaller than the sample SD. This difference provides limited descriptive evidence that the extreme slope estimates contributed to the sample SD, although it cannot be interpreted as the only reason for the observed dispersion. Taken together, these results indicate that a progressive within-session trend was not consistent across participants, while a small number of participants exhibited larger participant-specific trends whose sources cannot be determined from this analysis. Participant-specific offset trends may therefore warrant inspection in analyses requiring fine-grained temporal correspondence over extended recording periods.

Although the preceding regression analysis accounted for participant-specific offsets and linear within-session trends, swing-to-swing variability remained in the observed timestamp offsets. For each swing, we calculated a residual by subtracting the offset predicted by that participant's fitted regression line from the observed offset. The residual therefore represented the swing-specific deviation remaining after accounting for the participant-specific baseline and linear session trend. As a result, the residual mean was effectively 0 ms, the median absolute residual was 18.88 ms, and the SD was 111.68 ms (MAD-based robust SD 27.96 ms). The near-zero residual mean was expected because each participant-specific regression included an intercept and should therefore not be interpreted as evidence of negligible timing variability. The residual dispersion indicates that participant-specific starting offsets and fitted linear session trends did not fully account for the swing-to-swing variation in the observed cross-device timestamp offsets. Note that this residual variability characterizes variation in the correction values rather than post-correction synchronization error, as each per-swing offset was used for timestamp correction.

One potential contributor to this residual dispersion was timing uncertainty[56]. This uncertainty arose because the annotation identified a video frame, while the exact time of club-ball contact could be placed between two consecutive frame timestamps. The magnitude of this timing uncertainty varied with the frame rate of each video stream. For analyses using the front-view timestamps directly, the 30 fps frame rate contributed a maximum half-frame timing uncertainty of ± 16.7 ms. For streams recorded using the smartphone clock, the timestamp offset correction relied on both video streams, yielding a combined worst-case timing uncertainty of approximately ± 41.7 ms, comprising ± 25 ms for the 20 fps FPV anchor and ± 16.7 ms for the 30 fps front-view anchor. These theoretical bounds contextualized one possible contribution of the finite video frame rates but did not decompose or fully explain the empirical residual variability. Users performing analyses that rely on sub-frame temporal precision, such as instantaneous impact-time kinematics, should therefore account for these timing limitations, for example by using a time window around the impact event rather than a single timestamp.

Cross-Modal Timing Variability Characterization

Even when two streams from different sensors were timestamped using the same host-PC clock, this shared time base did not ensure identical acquisition timing or processing latency. Therefore, we additionally characterized the timing differences between the PNS-derived kinematic estimates and the front-view video using the same three analyses as in the previous section: the per-swing timing difference distribution, participant-specific within-session slopes, and residual swing-to-swing variability.

For the front-view video stream, we used the impact event timestamp annotated in the Data Processing section. Unlike the video stream, however, the PNS-derived kinematic stream did not provide a visually observable club-ball contact cue. Motivated by previous work reporting a salient wrist acceleration peak near the impact phase[64], we instead used the lead-hand synthetic acceleration peak as a consistently defined PNS kinematic landmark rather than as an estimate of the exact club-ball contact time. Because all participants performed right-sided swings, the LeftHand joint was used for every swing. The three-axis LeftHand synthetic acceleration signal was resampled onto a uniform 96 Hz time grid and filtered using a 10 Hz Butterworth low-pass filter (LPF). The magnitude of the filtered acceleration was then calculated using the Euclidean norm, and its global maximum was selected as the PNS landmark.

Of the 1,557 recorded swings, 1,540 containing both the PNS stream and a valid front-view contact annotation were included. For each swing, the timing difference was computed by subtracting the impact timestamp from the PNS landmark timestamp. The resulting timing difference distribution had a mean of -12.1 ms (median -3.50 ms, SD 133.78 ms, range -2549 to 1648.05 ms), indicating considerable temporal dispersion across swings. As shown in the wide range of the distribution, one of the prominent reasons was that the distribution contained several extreme values. Specifically, seven swings had absolute timing differences greater than 600 ms. Given the magnitude of these deviations, landmark localization failure is a plausible explanation for these cases, although it cannot be confirmed independently from the present analysis.

Using the same MAD-based robust SD described above, we obtained a robust SD of 53.04 ms, which was considerably smaller than the sample SD of 133.78 ms. This difference suggests that the sample SD was sensitive to the extreme observations. However, the robust SD remained non-negligible, indicating that the observed timing variability was not attributable solely to these extreme cases. Other possible contributors include the derivation and preprocessing of synthetic acceleration, localization

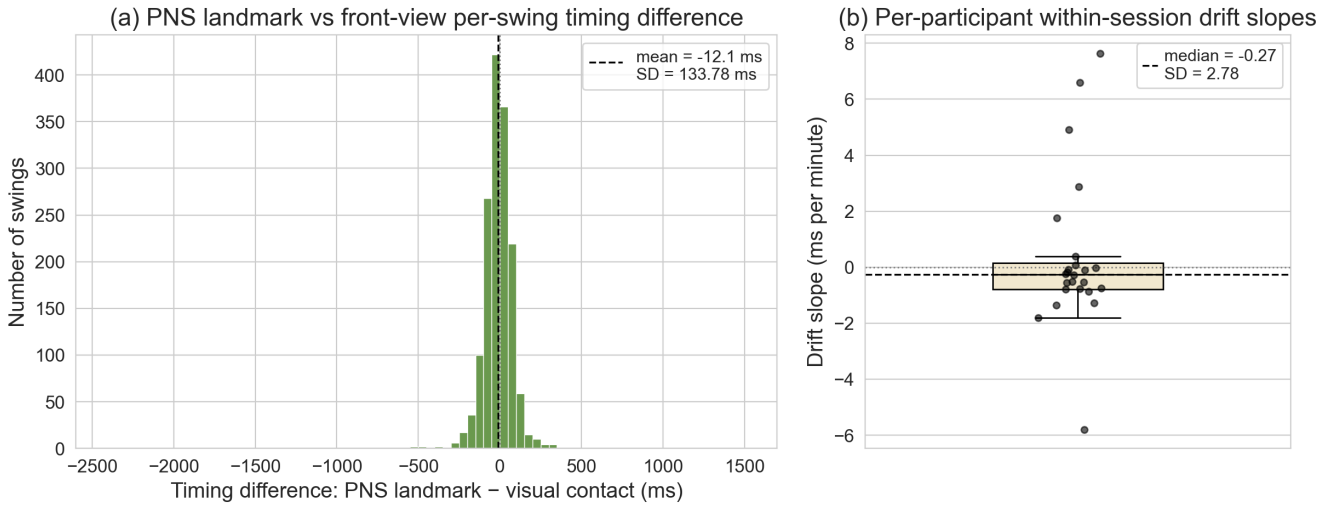

Figure 9. Timing variability between the PNS-derived kinematic stream and the front-view video. (a) Distribution of per-swing timing differences between the impact timestamp and PNS hand acceleration peak landmark timestamp. A dashed line indicates the mean timing difference. (b) Participant-specific within-session slopes obtained by regressing the timing difference against elapsed session time.

of the global acceleration peak on the discrete 96 Hz timestamps, timing uncertainty in the front-view annotation, manual annotation variability, and stream-specific acquisition or processing latency. Therefore, the small mean and median should not be interpreted as evidence that the PNS and front-view streams were temporally aligned. Instead, this analysis characterizes the observed timing-difference distribution and provides potential dataset users with an empirical reference when considering analyses that require fine-grained cross-modal temporal correspondence.

To further examine whether the observed timing difference changed progressively over each recording session, we fitted a participant-specific linear regression against elapsed session time[54,61]. As shown in Figure 9 (b), the resulting slopes had a mean of 0.34 ms/min, a median of -0.27 ms/min, and an SD of 2.78 ms/min. Although the slope estimates were centered near zero, they varied across participants. In particular, six participants (*P01*, *P05*, *P06*, *P07*, *P12*, and *P15*) had slope estimates outside Tukey's fences. Given that the MAD-based robust SD, 0.77 ms/min, indicates that the slope distribution was relatively concentrated around zero, the relatively large slope dispersion might have been affected by these extreme cases. Consistent with the offset correction characterization above, therefore, participant-specific temporal patterns may warrant inspection in analyses requiring fine-grained cross-modal correspondence over extended recording periods.

We additionally quantified swing-specific deviations between the PNS hand-acceleration landmark and visually annotated club-ball contact after accounting for each participant's baseline and linear session trend, using the same residual analysis as in the offset correction analysis section. The median absolute residual was 23.65 ms, and the MAD-based robust SD was 34.64 ms. Unlike the correction offset residuals, these residuals compared a PNS-derived kinematic landmark with a visually observed physical contact event and might therefore include variation in the relative biomechanical timing of the two events, together with landmark-localization variability, timing uncertainty in the visual annotation, and stream-specific acquisition or processing variability. Accordingly, the observed dispersion characterizes cross-modal timing correspondence and should not be interpreted solely as synchronization error.

Shot Performance Prediction Benchmark

Task definition

In this section, we evaluated supervised shot outcome prediction using five sensing input conditions recorded during a single golf swing. The benchmark was designed to provide compact and reproducible single-modality-based reference results for downstream reuse of the released data. Because each shot outcome was defined once per swing rather than per frame, each timeseries sensor stream was also independently summarized into one fixed-length swing-level feature vector. We evaluated five input conditions: PNS Lead-Hand, Head IMU, Pressure, FPV, and Front-View. Each condition was constructed from a distinct data modality. Three shot outcome targets were considered: ball speed, spin axis, and launch direction[65]. Ball speed was treated as a continuous regression target, while spin axis and launch direction were treated as three-class classification targets.

Table 4. Input feature definitions for the five input conditions.

Input condition	Frame-level representation	Swing-level summary	Dim.
PNS Lead-Hand	LeftHand synthetic acceleration $[a_x, a_y, a_z, \|\mathbf{a}\|]$ and address-relative rotation angle	RMS and range for the four acceleration channels; mean and range for the rotation angle	10
Head IMU	Raw acceleration $[a_x, a_y, a_z, \|\mathbf{a}\|]$ and raw angular velocity $[\omega_x, \omega_y, \omega_z, \|\omega\|]$	RMS and range for all eight channels	16
Pressure	Normalized left, right, and bilateral total pressure; left-load ratio; normalized left and right CoP coordinates	Mean and range for eight channels; left and right CoP path lengths	18
FPV	Strictly pre-impact optical flow represented by an 8-bin HOOF, its 8-bin temporal difference, and optical-flow magnitude	Mean and SD for all 17 channels	34
Front-View	Nine explicitly defined 2D geometric descriptors derived from COCO-17 body keypoints, as listed in the *Data Preprocessing* section.	Median and range for all nine descriptors	18

Note. Each native-rate stream was independently summarized into one fixed-length feature vector per swing.

Data preprocessing

To construct the benchmark dataset, we retained swings for which all five input conditions and all three shot outcome labels were available and valid. The same set of swings was used for every input condition so that performance differences were not attributable to differences in participant or swing composition. Of the 1,557 recorded swings, 1,088 swings satisfied these criteria. The remaining swings were excluded because one or more required sensor streams, impact timestamps, or shot outcome labels were unavailable. The data for each input condition was processed at its native sampling rate and summarized once over the corresponding swing interval. We did not resample the input streams onto a common time axis because interpolation across sparsely sampled signals could introduce artificial extrema that might adversely affect prediction.

The input feature extraction details including frame-level representations, temporal summaries, and final feature dimensions are provided in Table 4. For the *PNS Lead-Hand* condition, we used the 3-axis synthetic acceleration of the LeftHand joint, its magnitude, and an address-relative rotation angle. The rotation angle was computed from the relative orientation between each frame and the reference orientation estimated from the initial 200 ms of the swing. The four acceleration channels were summarized using their Root Mean Square (RMS) and range, while the rotation angle was summarized using its mean and range, yielding a 10-dimensional feature vector.

For the *head IMU*, we used the 3-axis acceleration and angular velocity together with their respective magnitudes, resulting in 8 frame-level channels. Each channel was summarized over the swing using its RMS and range, yielding a 16-dimensional feature vector. For the *Pressure* condition, we derived 8 features describing plantar-load distribution and CoP location: left-foot load, right-foot load, bilateral load, left-foot load ratio, and the two-dimensional CoP coordinates of each foot. Each feature was summarized using its mean and range, capturing the overall loading pattern and the extent of its variation during the swing. In addition, the cumulative CoP path length was calculated separately for the left and right feet to represent the total movement of the pressure center over the swing, which was not captured by coordinate ranges alone. This resulted in an 18-dimensional feature vector.

FPV features were extracted only from frames preceding the released impact timestamp to prevent visual information from the shot-result display from entering the predictor. The FPV representation comprised an 8-bin histogram of oriented optical flow (HOOF), its temporal difference, and optical-flow magnitude[66]. For the *Front-View*, 2D joint keypoints were extracted using the YOLO11n-pose COCO-17 model. Inspired by pose-based golf-swing metrics reported in prior work[65], we calculated 9 golf-specific metrics from the detected joint keypoints at each valid frame: *shoulder-line angle, pelvis-line angle, torso tilt, left and right elbow angles, right knee angle, normalized head displacement, normalized pelvis displacement,* and *stance ratio*. They were then summarized using their median and range, yielding an 18-dimensional feature vector.

Ball speed was used directly as a continuous regression target. In the final benchmark sample, ball speed had a mean of 37.666 m/s (SD 9.018 m/s; range 4.510–69.100 m/s). Spin axis was computed from the sidespin and backspin components and divided into hook or draw, straight, and fade or slice classes[67]. The class distribution was 214 (19.7%), 656 (60.3%), and 218 (20.0%), respectively. Launch direction was defined from the signed horizontal launch angle and divided into left, straight, and right classes[68], with 150 (13.8%), 489 (44.9%), and 449 (41.3%) swings, respectively. For both classification targets, the straight class was defined by the closed interval from -2° to $+2^\circ$.

Model configuration & Evaluation metrics

Because repeated swings from the same participant might expose participant-specific characteristics to both training and test sets, we used nested group k-fold cross-validation with participant ID as the grouping variable. In this configuration, the participants were assigned to four outer folds, with each participant appearing in the test set exactly once and no participant

Table 5. Shot outcome prediction performance of the no-sensor baseline and five input conditions.

Condition	Ball Speed			Spin Axis			Launch Direction		
	R^2	RMSE	MAE	Macro-AUROC	Bal. Acc.	Macro F1	Macro-AUROC	Bal. Acc.	Macro F1
No-sensor baseline	−0.024	9.074	7.221	0.500	0.333	0.251	0.500	0.333	0.208
PNS	**0.054**	**8.731**	**6.325**	0.548	**0.401**	0.337	0.464	0.319	0.277
Head IMU	−0.204	9.486	7.488	0.566	0.360	0.315	0.508	0.353	0.334
Pressure	−0.155	9.663	7.892	0.523	0.342	0.283	0.506	0.330	0.295
FPV	−0.055	9.211	7.432	0.580	0.387	**0.370**	**0.575**	0.390	**0.356**
Front-view	0.014	8.892	7.024	**0.615**	0.380	0.368	0.542	**0.398**	0.335

Note. Values are means across four outer folds. Greater values are better for R^2, macro-AUROC, balanced accuracy, and macro F1, while smaller values are better for RMSE and MAE. The best value in each column is in **bold**.

appearing in both the training and test partitions. Within each outer training set, participant-level three-fold cross-validation was used for model family and hyperparameter selection.

For ball speed regression, we considered Ridge regression[69] and XGBoost regression[70]. For spin axis and launch direction classification, we considered multinomial logistic regression and XGBoost classification. Ridge regularization and logistic regression were evaluated using five logarithmically spaced candidates from 10^{-4} to 10^{4}. Five conservative XGBoost configurations were evaluated, with maximum tree depths ranging from 2 to 4. The model family and hyperparameter configuration with the best performance according to the prespecified inner-validation criterion was refitted on the full outer training set and evaluated once on the corresponding outer test fold. Within each outer fold, missing-value imputation and standardization were fitted using only the outer training set, and constant features were identified and removed based on the same training data. These preprocessing steps were then applied unchanged to the held-out outer test set. Class weights were calculated from the outer-training labels and used only during model fitting. The same participant-level outer folds were used across all input conditions and prediction targets.

To assess whether the sensing inputs provided predictive information beyond the outcome distribution of the training data, we defined a no-sensor baseline using only the outer training labels. For the no-sensor regression baseline, the mean ball speed of the outer training set was assigned to every outer test swing. For the classification baseline, the empirical outer training class proportions were used as probability predictions, and the majority training class was used as the categorical prediction. The no-sensor baseline used neither inner cross-validation nor feature preprocessing.

For the ball speed regression, we reported the coefficient of determination (R^2), RMSE, and mean absolute error (MAE). For the spin axis and launch direction classification, we reported macro-averaged one-vs-rest area under the receiver operating characteristic curve (macro-AUROC)[71], balanced accuracy, and macro F1. Macro-AUROC and macro F1 were calculated as unweighted averages of the corresponding class-specific scores, whereas balanced accuracy was calculated as the mean recall across classes.

Benchmark Results

Table 5 presents the results of the no-sensor baseline and the five input conditions. Among them, PNS Lead-Hand yielded the greatest observed point estimate for ball speed R^2 (0.054 ± 0.159) and the smallest RMSE (8.731 ± 1.353 m/s) and MAE (6.325 ± 1.213 m/s). PNS Lead-Hand and Front-view produced better aggregate point estimates than the no-sensor baseline for all three ball-speed metrics, whereas the other three conditions did not. These results are consistent with the report from prior work that joint kinematic information including wrist velocity and X-factor is associated with ball velocity[8].

For spin-axis classification, the greatest observed macro-AUROC was obtained using Front-view features (0.615 ± 0.077), while PNS Lead-Hand yielded the greatest balanced accuracy (0.401 ± 0.064) and FPV yielded the greatest macro F1 (0.370 ± 0.016). For launch-direction classification, FPV yielded the greatest observed macro-AUROC (0.575 ± 0.073) and macro F1 (0.356 ± 0.031), while Front-view yielded the greatest balanced accuracy (0.398 ± 0.063). Head IMU, FPV, and Front-view produced greater aggregate point estimates than the no-sensor baseline for all three launch-direction metrics.

The direction and magnitude of the differences varied across sensing inputs, prediction targets, and metrics. Overall predictive performance also remained limited, particularly for ball speed prediction. Nevertheless, for each of the three prediction targets, at least two sensing inputs yielded better aggregate point estimates than the no-sensor baseline across all reported metrics. The input conditions associated with the best observed performance also differed across targets and metrics. Taken together, these findings can be interpreted as limited descriptive evidence that the released MultiSenseGolf streams can support reproducible shot outcome analysis under the present benchmark protocol and that the relative utility of individual

Table 6. Assessment of potential selection bias using participant-clustered Wald tests.

Variable set	No. of variables	Wald statistic	Bootstrap p
Shot outcomes	3	14.8894	0.0542
Input features	10	23.8678	0.5506
Combined set	13	57.3597	0.3626

Note. p-values were calculated using 4,999 participant-level wild-bootstrap replicates.

input conditions may vary with the analysis objective.

However, because the present benchmark used only the single-modality-based input conditions, it did not directly test cross-modal complementarity or fusion benefits. More broadly, the analysis was designed as a compact benchmark to provide a reproducible baseline for downstream dataset reuse, rather than as an optimized prediction system. We therefore suggest several possible extensions to this benchmark. For example, future users could retain the full timeseries instead of summarized input features, allowing them to apply models that capture how the signals change over time within a swing, such as Long Short-Term Memory (LSTM) networks or Transformers. Another direction is to combine complementary inputs, such as joint kinematics with foot pressure or video, and to include gaze and depth data that were not used in the present benchmark.

Effect of Excluding Swings with Missing Data

The benchmark included 1,088 swings with all required input and target data available, while excluding 469 swings with missing required data. This selection could introduce bias if data availability was associated with swing motion or shot outcomes, resulting in a benchmark sample with a different composition from the full dataset. Such differences could affect the observed predictive performance and limit its generalizability to the broader set of recorded swings. To examine whether benchmark inclusion was associated with differences in the observed data or predictive performance, we conducted two additional analyses: a comparison of the 1,088 included swings with the 469 excluded swings, and an evaluation of the benchmark models on expanded test sets containing excluded swings.

For the mean comparisons, we used an adapted generalized estimating equation (GEE) approach that accounted for dependence among repeated swings from the same participant, following the parameter-homogeneity principle of Chen and Little[72]. We assessed mean differences jointly within three sets of variables: three shot outcomes, ten input features comprising two features per modality, and all thirteen variables combined. These comparisons used the available observations for each variable without imputation, with p-values calculated using 4,999 participant-level wild-bootstrap replicates. The joint comparisons detected no statistically significant differences in observed means between included and excluded swings at the nominal 0.05 level (Table 6). These results indicate that the observed data did not provide sufficient evidence of differences in average shot outcomes or selected input features between included and excluded swings. However, these results should not be interpreted as demonstrating equivalence between included and excluded swings.

The performance sensitivity analysis assessed how benchmark performance changed when usable excluded swings were added to the evaluation sets. While retaining the original training samples, model settings, preprocessing procedures, and participant folds, we added 37-295 previously excluded swings to the test sets, depending on the availability of the required modality and target. Specifically, a swing was eligible for evaluation if its input modality met the original feature-extraction requirements and its corresponding prediction target was observed, even if data from other modalities or targets were missing. Each added swing was evaluated using the fold model trained without that participant. Performance was recalculated from out-of-fold predictions for both the original and expanded evaluation sets. Across modalities, Ball Speed RMSE changed by -0.272 to $+0.151$ m/s, corresponding to relative changes of -2.81% to $+1.54\%$ from the original estimates. Across the two classification tasks, balanced accuracy changed by -1.83 to $+1.05$ percentage points, corresponding to relative changes of -5.56% to $+2.99\%$. These results imply that adding previously excluded swings produced limited changes in aggregate predictive performance, although the largest decreases in balanced accuracy exceeded the largest increases in both classification tasks.

Overall, the GEE comparisons did not detect statistically significant joint mean differences between included and excluded swings. The performance sensitivity analysis also showed limited changes in aggregate predictive performance when usable excluded swings were added, suggesting that the reported performance was relatively stable to this expansion. Together, these findings provide an assessment of observable selection differences and their potential implications for benchmark performance. However, the possibility of selection bias cannot be fully ruled out because the analyses could not assess unobserved values or predictive performance for swings lacking the required inputs or targets.

Usage Notes

Potential Applications

We propose several potential golf-related applications that can be developed by leveraging the dataset or expanding it with further annotations. First, the dataset can support biomechanical analyses that jointly examine cross-skill and cross-modality relationships within the same swing instances. For example, users can quantify biomechanical tendencies by comparing swing-specific kinematics such as differences in pelvis and thorax rotation timing[73], or weight-shift patterns[42]. Cross-modality relationships can also be studied by examining how wearable signals correspond to kinematic variables, such as estimating lower-body events from foot pressure patterns. In addition, the shot measurement data such as ball speed and carry distance enables supervised shot outcome prediction tasks using multimodal sensor inputs[29], allowing users to evaluate how different modality combinations contribute to predictive accuracy.

Users can also extend the dataset by creating additional annotations on the existing recordings. For instance, manually labeling swing events including address and backswing enables phase-level analysis and swing event detection[21]. Furthermore, annotation can be extended to incorporate expert-level analysis, supporting Action Quality Assessment (AQA)[74] and constructing intelligent coaching systems[15]. By leveraging the front-view video, golf professionals can assign proficiency scores or provide textual feedback on motion correction. This data allows the training of models not only to assess the overall swing movement quality of users from sensor inputs[23], but also to develop a natural language-based coaching pipeline with LLMs.

Dataset Limitations

This section outlines specific limitations of the MultiSenseGolf dataset regarding hardware characteristics and experimental design. These constraints are discussed for dataset users to ensure appropriate data utilization and to suggest potential directions for future research.

First, the PNS streams should be interpreted as IMC-derived joint kinematic estimates rather than raw IMU measurements. No raw accelerometer, gyroscope, or magnetometer measurements from the 17 body-worn PNS IMUs are included in the dataset. The provided PNS angular velocity and acceleration streams are likewise derived from the estimated joint orientations and positions rather than measured directly by these units. The orientation and position estimates can support applications emphasizing temporal movement patterns and inter-joint coordination, including pose reconstruction, movement-pattern characterization, and swing-phase analysis. However, these estimates should not be treated as independently validated ground truth for applications requiring metrically precise joint positions or direct sensor-level inertial measurements. In particular, the joint positions should be treated carefully. Because they are reconstructed through FK, errors in orientation estimates and sensor displacement relative to the underlying body segments may propagate and accumulate along the kinematic chain, resulting in potentially larger positional errors at joints farther from the root. Our FK sensitivity analyses, based on the assumption that the orientation errors were similar to the vendor-specified attitude accuracy and task-based joint-angle RMSEs reported in optical-reference validation studies of PNS, yielded RSS positional sensitivities smaller than approximately 150 mm[35,36], as described in the Technical Validation section.

Second, users should be aware of the inherent drift, noise, and data loss caused by the wearable hardware used in this study. The 3D motion capture may contain residual errors from IMU drift or sensor displacement caused by centrifugal forces during high-speed swings, despite real-time monitoring and recalibration. Additionally, the wired FPV setup, chosen to minimize sensor mass and packet loss, occasionally caused connectivity instability during dynamic movements, leading to recording failures in some trials. Users should therefore filter swings to retain only those with all required modalities for multimodal analyses.

Third, several modalities in this dataset were subject to temporal-resolution and signal-fidelity limitations that users should consider when designing analyses. As described in Background & Summary, the fast dynamics of the swing were covered by the high-frequency IMC streams, whereas the lower-frequency modalities were designed to capture more slowly varying contextual information. Accordingly, the FPV video, recorded at 20 fps, was sufficient for capturing egocentric visual context but was not suited for frame-level analyses of fast-moving objects such as the clubhead. We suggest that potential users supplement the onboard IMU signals from the FPV stream when performing analyses requiring sub-frame temporal precision. Similarly, the insole pressure data, recorded at 10 Hz, was sufficient for slowly varying signals such as backswing weight shifts, but did not resolve rapid kinetic events such as the peak GRF or the instantaneous weight transfer at impact. This sampling rate reflected a trade-off in the wireless transmission bandwidth of the ESP32-based readout module, which had to relay a total of 480 sensing points in real time. We therefore suggest that analyses of such events instead rely on the 96 Hz motion capture stream. In addition, the pressure data was provided as raw ADC values rather than calibrated absolute pressure values. Although these are sufficient for the motion estimation tasks by capturing relative contact patterns and weight distribution, they should not be used as absolute foot pressure or GRF measurements without additional calibration.

Fourth, two experimental factors might have influenced the swing performance of the participants during data collection. First, wearing sensors introduced physical encumbrance that could slightly constrain swing motion or body balance. Although

we selected lighter hardware and instructed participants to prioritize swing form over shot performance, post-hoc surveys and interviews indicated that participants still experienced some unnaturalness. Future research can leverage the Comfort Rating Scale and qualitative feedback in this dataset to explore ergonomic sensor configurations. Second, the large number of swings required for the Professional group (a minimum target of 100 swings per session) might have introduced physical fatigue that affected kinematics and postural consistency, particularly toward the end of a session. Although several fatigue mitigation strategies were applied during data collection such as free-form rest and recording checks every 10 swings, residual fatigue effects can still confound the interpretation of intra-participant variance. Therefore, users should account for fatigue as a potential confounder. A detailed characterization of fatigue effects on swing kinematics also remains for future work.

Finally, the experimental setup introduced controlled conditions that could limit the diversity of the collected data. First, the dataset was collected in a controlled indoor golf center, which ensured privacy and enabled efficient use of built-in equipment. However, this setting lacked the variability introduced by outdoor factors, such as irregular terrain, wind, and illumination variation. Such factors can affect individual sensor modalities differently, including degraded video quality under changing illumination and altered foot pressure patterns on uneven ground. These modality-specific changes may lead to distribution shifts, thereby limiting the generalization of models trained solely on the controlled indoor data to real-world field environments. Second, the diversity of swing mechanics was limited by the single-club (7-iron) protocol, which we adopted to prioritize data quality. Real-world golf involves a diverse range of clubs (e.g., drivers, woods, putters), each with distinct swing kinematics. For example, driver swings typically exhibit a wider stance and faster upper-body rotational velocities than iron swings. We suggest that future work aiming to develop more generalized models use the dataset as a pretraining source and then adapt models to broader settings through domain adaptation techniques such as transfer learning[75].

Overall, MultiSenseGolf provides a publicly available multimodal dataset comprising 1,557 golf swings from 24 participants across Beginner, Intermediate, and Professional groups. By linking joint kinematic estimates, bilateral foot pressure recordings, and egocentric and external video with shot outcomes at the swing level, it provides complementary observations for studying relationships between swing motion and performance. The inclusion of participant metadata supports comparisons across skill levels, while the release of data processing, analysis, and benchmark code facilitates reproducible research. The technical analyses help users interpret the dataset's measurement and timing characteristics, and the single-modality benchmark provides reference results showing variation in observed predictive performance across inputs and prediction targets. Within the scope of the participant sample, controlled indoor 7-iron protocol, and documented measurement limitations, MultiSenseGolf provides a reusable basis for further golf swing analysis and for the development and evaluation of sensor-based shot outcome prediction models.

Code availability

The custom code developed for data processing, analysis, and benchmark evaluation is publicly available at the following GitHub repository: https://github.com/Oh-Minwoo/MultiSenseGolf.git. The repository provides a tutorial to assist users in loading each data format (HDF5, MP4, PNG), executing preprocessing pipelines, and visualizing multimodal streams. The statistical analysis of carry distance across participant characteristics reported in this study is also included as a separate script. Furthermore, we release the full source code for the benchmark test, covering the entire pipeline from feature extraction to model training and evaluation, along with example configuration files.

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Author contributions

M.O. prepared sensors and data streaming frameworks, conducted data collection, post-hoc processing and technical validation, and wrote the manuscript. M.S. provided critical feedback for research direction shaping, supported software development, and reviewed the manuscript. K.K. participated in the design and coordination of the study, and reviewed the manuscript. C.L. prepared sensors, and supported data collection. G.K. reviewed the manuscript. H.C. designed and developed the custom sensor hardware, and prepared a figure in the manuscript. J.D. and D.R. provided high-level comments on the manuscript. S.K. supervised the project administration, provided overall guidance throughout the study, and reviewed and approved the final manuscript.

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